in the fuel a high fission rate dependence. The doppler broadening also increases capture in the control rods. These latter effects normally dominate over the increase in f, so that k_∞ decreases slightly with increasing reactor power and temperature. This is referred to as a *negative temperature coefficient*. In a BWR a negative temperature coefficient is also caused by the bubbles of steam (*void*) occurring in the moderator, which reduces the thermalization of neutrons while increasing neutron leakage rate. In BWRs the effect of the void normally dominates over doppler broadening in limiting the reactor power. As the reactor tends to keep the void fraction and fuel temperature constant, the power level of a BWR (and to a smaller degree also of a PWR, where mostly the average fuel temperature is affected) can be adjusted by varying the speed of the coolant circulation pumps at a constant control rod setting.

These are important safety features of a reactor. If k increases, so do the fission rate and temperature. If the temperature coefficient was positive, k would increase further, leading to further temperature increase, etc. However, with a negative temperature coefficient the reactor controls itself: an increase in power (and thus temperature) decreases k, which tends to limit the power increase and vice versa. The temperature coefficient is usually given in $\%\Delta k/k$ per °C.

19.7. NEUTRON LEAKAGE AND CRITICAL SIZE

All practical reactors have some leakage of neutrons out of the reactor core. This leakage Λ is approximately described by the fast and thermal leakage factors Λ_f and Λ_{th} respectively:

$$\Lambda = \Lambda_f \Lambda_{th} \tag{19.15}$$

The effect of this leakage is included in Figure 19.6, and results in the number of neutrons in the second generation being reduced, from the 112 calculated previously for k_∞, to $N_2 = 101$. Thus $k_{eff} = 1.01$ and $\Lambda = 0.9$, i.e. 10% of the neutrons are lost through leakage. The example in Figure 19.6 refers to a small air cooled and graphite moderated natural uranium research reactor. In nuclear power stations k_∞ is much larger due to the large η-values (because of the enriched fuel needed to sustain operation for long periods at high power between refuelings), while the leakage is much smaller, $\Lambda \geq 0.97$ (e.g. see Table 19.4).

The leakage is a function of the geometrical arrangement of reactor core and reflector and of the average distance the neutron travels after formation until it causes a new fission:

$$\Lambda_f = e^{-B^2\tau} \tag{19.16}$$

$$\Lambda_{th} = \left(1 + B^2 L^2\right)^{-1} \tag{19.17}$$

where B^2 is the *geometrical buckling*, L the *thermal diffusion length*, and τ is referred to as the *neutron* (or *Fermi*) *age*.

Combining these equations gives the *critical equation*:

$$k_{\text{eff}} = k_\infty \, \mathrm{e}^{-B^2\tau} / \left(1 + B^2 L^2\right) \approx k_\infty / \left(1 + B^2 \left(L^2 + \tau\right)\right) \tag{19.18}$$

The quantity $L^2 + \tau$ is usually denoted by M^2 and is known as the *migration area* and M as the *migration length*:

$$M^2 = L^2 + \tau \tag{19.19}$$

Hence for a large thermal critical reactor one can write

$$k_{\text{eff}} = k_\infty / \left(1 + B^2 M^2\right) = 1 \tag{19.20}$$

The thermal diffusion length L is calculated from

$$L^2 = L_{\text{m}}^2 (1 - f) \tag{19.21}$$

where L_{m} is the diffusion length in the pure moderator. L_{m}^2 and τ are given in Table 19.3 for various moderator materials.

The geometrical buckling B^2 depends on the neutron flux distribution in the reactor. This distribution in turn depends on the general geometry of the assembly, including boundary conditions. The calculation of the buckling for heterogeneous reactors is quite complicated but for homogeneous and bare (no reflector) reactors the following approximate simple relations hold:

$$B^2 = \pi^2 r^{-2} \qquad \text{(sphere)} \tag{19.22}$$

$$= 3\,\pi^2 a^{-2} \qquad \text{(cube)} \tag{19.23}$$

$$= (\pi/h)^2 + (2.405/r)^2 \qquad \text{(cylinder)} \tag{19.24}$$

$$= 33\, h^{-2} \qquad \text{(cylinder; height = diameter)} \tag{19.25}$$

where r is the radius of the sphere and cylinder, a the side of the cube, and h the cylinder height (which is also the diameter of the cylinder in eqn. 19.25).

It is seen that k_{eff} in (19.18) increases with decreasing buckling, and, because $B \propto 1/r$ (19.22), with increasing size of the reactor. This is a result of the neutron production k_∞ being a volume effect (proportional to r^3 for a sphere), whereas leakage is a surface effect (proportional to r^2). For each reactor there is a minimum *critical size* ($k_{\text{eff}} = 1$) below which the surface to volume ratio is so large that neutron leakage is sufficient to prevent the fission chain reaction.

The smallest critical sizes are obtained for *homogeneous systems* of pure fissile nuclides with maximum neutron reflection. For neutrons with the fission energy spectrum, the critical mass of a metallic sphere of pure ^{235}U is 22.8 kg, that of ^{233}U is 7.5 kg, and that of ^{239}Pu is 5.6 kg, assuming a 20 cm uranium metal neutron reflector. For fission by thermal neutrons the smallest critical size of a spherical *homogeneous aqueous solution* of $^{235}UO_2SO_4$ without reflector requires 0.82 kg of ^{235}U in 6.3 l of solution. The corresponding figures for ^{233}U are 0.59 kg in 3.3 l, and of ^{239}Pu, 0.51 kg in 4.5 l.

Homogeneous solutions of fissile nuclides are produced in the reprocessing of spent fuel elements, where care must be exercised that the critical size is not exceeded in any equipment or container in order to prevent an accidental chain reaction. Several such accidents have occurred in the past in fuel production and reprocessing plants in which very high doses were received by nearby personnel even though the duration of the chain reaction usually was very short and a violent explosion never occurred.

For heterogeneous reactors it is more difficult to quote comparable simple values for critical size. These have to be calculated by numerical methods or determined empirically for each particular reactor configuration.

19.8. REACTOR KINETICS

The mean lifetime θ for a neutron in a reactor is the time it takes on the average for the neutrons to complete one loop in the neutron cycle. In thermal reactors θ is 10^{-3}–10^{-4} s due to the comparatively low speed of thermalized neutrons and average distance in the moderator covered by random walk when travelling from and to a fuel pin. For each loop the number of neutrons is multiplied by a factor k_{eff}. Since one neutron is used for maintaining the chain reaction, the neutrons in the reactor change with time according to

$$\mathrm{d}N/\mathrm{d}t = N\left(k_{\text{eff}} - 1\right)/\theta + K \tag{19.26}$$

where N is the total number of neutrons in the reactor, and K is the contribution from any constant neutron source present. Solving this equation, we obtain

$$N = N_0\, \mathrm{e}^{\left(k_{\text{eff}}-1\right)t/\theta} + K\,\theta\left(1 - \mathrm{e}^{\left(k_{\text{eff}}-1\right)t/\theta}\right)\Big/\left(1 - k_{\text{eff}}\right) \tag{19.27}$$

where N_0 is the number of neutrons at $t = 0$. We can distinguish three different cases:

(i) $k_{\text{eff}} < 1$ and $t \gg \theta$: the reactor is subcritical,

$$N(t) = K\,\theta\left(1 - k_{\text{eff}}\right)^{-1} \tag{19.28}$$

The initial number of neutrons, N_0, has disappeared and the number of neutrons is directly related to the constant neutron source. The reactor acts as a neutron amplifier, with amplification increasing with k_{eff}.